

Week 3 Report

PENTAS HUB Internship Program

Project Title

Diabetes & Heart Disease Risk Predictor Using Hybrid Machine Learning Model

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Week Number: Week 3

Domain: Machine Learning

Duration: Week 3

1. Introduction

During Week 3 of the internship, the primary objective was to enhance the disease prediction system by implementing a hybrid machine learning approach. Multiple classification algorithms were trained and compared to improve the prediction accuracy for both Diabetes and Heart Disease. A Hybrid Voting Classifier was developed by combining Logistic Regression, Random Forest, and Support Vector Machine (SVM) models.

2. Objectives

- Build a Hybrid Machine Learning model.
 - Train multiple classification algorithms.
 - Compare model performance.
 - Improve prediction accuracy.
 - Save trained models for future prediction.
 - Prepare the project for deployment.
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3. Tasks Completed

Dataset Preparation

- Used the cleaned Diabetes dataset.
- Used the cleaned Heart Disease dataset.

- Verified dataset consistency.
- Loaded datasets into Python.

Model Development

The following machine learning models were trained:

- Logistic Regression
- Random Forest
- Support Vector Machine (SVM)

Hybrid Model

A Hybrid Voting Classifier was implemented by combining the predictions of:

- Logistic Regression
- Random Forest
- SVM

The hybrid model uses Soft Voting to improve prediction performance.

4. Technologies Used

- Python
 - Pandas
 - NumPy
 - Scikit-learn
 - Joblib
 - Google Colab
 - GitHub
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5. Files Developed

Source Files

- train_models.py
- hybrid_model.py
- evaluate_models.py
- predict.py

Datasets

- diabetes_cleaned.csv

- heart_cleaned.csv

Model Files

- logistic_regression_diabetes.pkl
 - random_forest_diabetes.pkl
 - svm_diabetes.pkl
 - hybrid_diabetes_model.pkl
 - logistic_regression_heart.pkl
 - random_forest_heart.pkl
 - svm_heart.pkl
 - hybrid_heart_model.pkl
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6. Evaluation Metrics

The trained models were evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score

The Hybrid Voting Classifier achieved improved prediction performance compared to individual models by combining their strengths.

7. Results

- Successfully trained multiple machine learning models.
 - Successfully created Hybrid Voting Classifier models.
 - Saved trained models as `.pkl` files.
 - Compared model performance using evaluation metrics.
 - Generated prediction models for Diabetes and Heart Disease.
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8. Challenges Faced

- Selecting suitable machine learning algorithms.
 - Combining multiple models into a hybrid classifier.
 - Managing model serialization using Joblib.
 - Evaluating and comparing multiple models efficiently.
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9. Solutions Implemented

- Used Scikit-learn's VotingClassifier to build the hybrid model.
 - Applied a train-test split for model validation.
 - Evaluated models using multiple performance metrics.
 - Saved trained models for reuse during prediction.
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10. Learning Outcomes

During Week 3, I learned:

- Hybrid Machine Learning concepts.
 - Ensemble learning using Voting Classifier.
 - Model comparison techniques.
 - Performance evaluation using Accuracy, Precision, Recall, and F1 Score.
 - Model serialization using Joblib.
 - Organizing machine learning projects using GitHub.
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11. Conclusion

Week 3 focused on enhancing the disease prediction system through a hybrid machine learning approach. Multiple classification models were trained, evaluated, and combined into a Hybrid Voting Classifier to improve prediction performance. The trained models were successfully saved and integrated into the prediction workflow. This work establishes a strong foundation for Week 4, where transformer-based models will be incorporated for processing medical text and patient notes.

12. Future Work

- Integrate transformer models for medical text analysis.
- Process patient notes using BERT.
- Improve explainability of predictions.
- Develop a Retrieval-Augmented Generation (RAG) chatbot.
- Deploy the final application using Streamlit in Week 5.